



BROWN

Attention-Based BiLSTM for Sentiment Classification

CSCI 1470 Deep Learning Final Project – Attention Is All We Need

Team Members: Xiaoqing Yao, Linzhao Zhang, Lin Zhou



1. INTRODUCTION / MOTIVATION

Sentiment analysis of short texts, such as movie reviews, remains a significant challenge in Natural Language Processing due to limited contextual information and sparse semantic features. Traditional machine learning approaches often rely heavily on complex manual feature engineering and struggle to capture deep semantic dependencies. In this project, we implement an Attention-based Bidirectional Long Short-Term Memory (AB-BiLSTM) neural network to address these limitations. By integrating a BiLSTM architecture with an attention mechanism, our model can capture long-range dependencies from both forward and backward directions while dynamically weighing the most informative words in a sentence. We evaluate our implementation on the IMDB dataset, aiming to demonstrate that the synergy between bidirectional recurrent networks and attention layers provides superior performance in capturing the qualitative parts of a text without the need for external linguistic resources.

Attention-based BiLSTM Neural Networks for Sentiment Classification of Short Texts

February 2018

DOI:10.22323/1.300.0014

License · CC BY-NC-ND 4.0

Conference: Information Science and Cloud Computing

4. DISCUSSION

Attention helps the model focus on sentiment-bearing words.

BiLSTM + Attention slightly outperformed iLSTM + Attention in our setup.

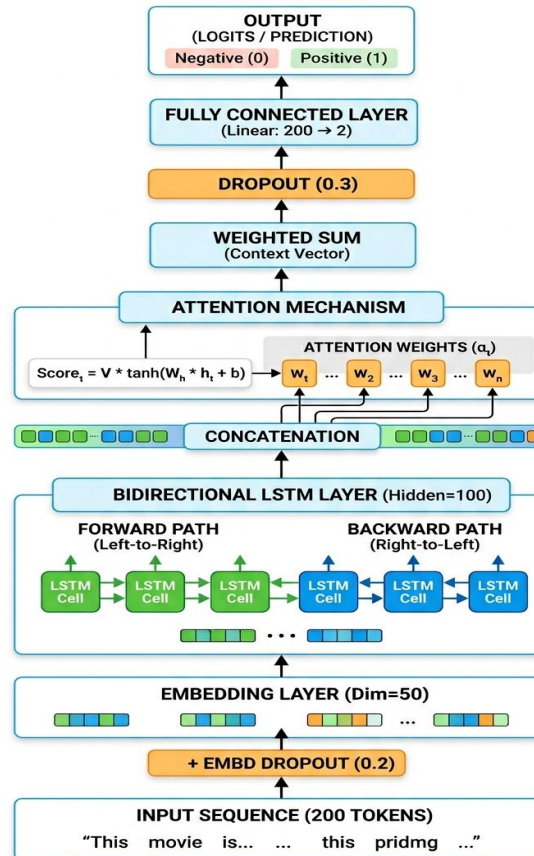
For reviews around 200 words or fewer, the simpler attentive BiLSTM may already capture enough context.

The more complex BiLSTM + Attention model may have been more prone to overfitting on this dataset.

Training accuracy > 92%, while validation accuracy stayed around 89% or lower.

2. METHODOLOGY

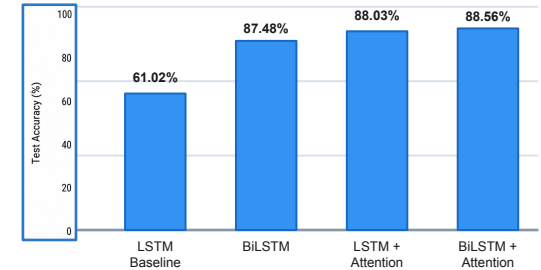
Model Pipeline (BiLSTM + Attention)



Dataset: IMDB 50K (Kaggle) This project utilizes the IMDB 50K movie review dataset (via Lakshmi BN, Kaggle), a benchmark for binary sentiment classification. It contains 50,000 polarized reviews originally collected by Stanford researchers, featuring a balanced 1:1 ratio of positive and negative samples (25,000 each). To ensure label reliability, the dataset only includes extreme ratings (≥ 7 or ≤ 4).

3. QUANTITATIVE RESULTS

Test Accuracy Comparison



Model	Test Accuracy(200 token)	Test Accuracy(50 token)
LSTM Baseline	61.02%	78.90%
BiLSTM	87.48%	78.32%
LSTM + Attention	88.03%	78.72%
BiLSTM + Attention	88.56%	79.62%



Best observed test accuracy:
88.56% (BiLSTM + Attention).

Attention improved performance by roughly 27% over the LSTM baseline.



5. FUTURE WORK

Hyperparameter Optimization: Further tune hidden dimensions and learning rates to find the optimal balance between model capacity and speed to prevent overfitting.

Architectural Evolution: Extend the current BiLSTM+Attention framework to Multi-Head Attention or Transformer-based architectures for improved long-range dependency modeling.